

# RÚBEN ARAÚJO

## RESEARCH EXPERIENCE (Biomedical / Clinical)

**Researcher – Instituto Superior de Engenharia de Lisboa (ISEL)** Oct 2017 – Present | Lisbon, Portugal

- Led and supported biomedical and clinical research involving patient-level clinical, laboratory, therapeutic and omics data, biospecimen handling, sample preparation, protocol optimization, biomarker analysis and multidisciplinary research workflows. Developed analytical tools and workflows for large-scale data mining, biomarker discovery and machine-learning-based disease classification and outcome prediction.

**Researcher – Comprehensive Health Research Center (CHRC)** Nov 2021 – Present | Lisbon, Portugal

- Integrated multidisciplinary clinical and biomedical research groups, contributing to biomarker-focused studies and collaborative scientific activities.

**Studentship – Fundação para a Ciência e a Tecnologia (FCT)** Nov 2021 – Nov 2025 | Lisbon, Portugal

- Doctoral research focused on ICU-based biomarker discovery and clinical outcome stratification.

**Principal Investigator – Instituto Português da Face (IPF)** 2020 | Lisbon, Portugal

- Led clinical research on TMJ tissue regeneration and 3D surgical planning, including sample/data processing, statistical analysis, and scientific publication.

## CLINICAL DATA, AI & BIOMARKER ANALYSIS

- Analysis of large, longitudinal clinical, laboratory, therapeutic and omics datasets for disease classification, prognosis and outcome prediction.
- Clinical biomarker discovery using machine learning, statistical modelling, data mining and multivariate analysis.
- Integration and interpretation of patient-level clinical and molecular data from hospital-based research cohorts.
- Development of Python-based analytical tools for automated data processing, biomarker screening and large-scale combination analysis.
- Development of AI-assisted research tools, including LLM-supported interfaces for data interpretation and analytical workflow support.

## RESEARCH PROJECTS

- Participation in 8 funded research projects as Researcher, Principal Researcher and Founder, with combined funding across projects of ~€1 million.
- Domains:** clinical metabolomics, biomedical devices, infection diagnostics, critical care, TMJ regenerative solutions, nephrology, machine learning applied to clinical medicine.
- Institutions involved:** NMS, ISEL, CHULC, IST, IPST, IPL, and Fundação Champalimaud.
- Experience with FCT, Portugal 2020/FEDER and multidisciplinary biomedical research frameworks, including clinical data analysis, computational workflows, biomarker development and translational research.

## PUBLICATIONS & SCIENTIFIC OUTPUT

- 30+ peer-reviewed publications, including 9 first-author articles.
- Quartiles: 13 Q1, 13 Q2, 2 Q3, 1 Q4.
- Topics include biomarkers, clinical data analysis, critical care mortality, nephrology, transplantation, infection, omics and machine learning.

## MEDICAL RESEARCH INNOVATION

- National patent application: PT120761 (2025), Infection Diagnosis.
- International patent contribution: PCT/PT2025/050006, Allograft Rejection Prognosis.
- Experience supporting patent documentation, prior-art positioning, and research-to-application discussions in biomedical projects.

## EDUCATION

**PhD in Biomedicine – NOVA Medical School (NMS)** 2021 – 2026 | Lisbon, Portugal

**MSc in Biomedical Engineering – ISEL** 2017 – 2019 | Lisbon, Portugal

**MSc in Mechanical Engineering (Maintenance & Production) – ISEL** 2007 – 2011 | Lisbon, Portugal

**BSc in Mechanical Engineering – ISEL** 2002 – 2007 | Lisbon, Portugal

## ANALYTICAL & TECHNICAL TOOLS

- Python / R / SPSS / SQL
- Power BI / Tableau
- Machine Learning & Data Mining
- Generative AI / LLMs / AI-assisted Analytics
- Clinical & Omics Data Analysis
- Reproducible Analytical Workflows
- Scientific Reporting & Documentation

## SCIENTIFIC INFRASTRUCTURE

- Scientific Project Management
- Clinical Research Coordination
- Translational Biomedical Research
- Omics-Based Biomarker Development
- EU & National Funding Frameworks
- Multidisciplinary Team Interface

## LANGUAGES

- Portuguese – Native
- English – C1 (IELTS + Cambridge)
- Norwegian – B1 (Aftenskolen)
- Spanish – B1 (Instituto Cervantes)

## ACADEMIC LEADERSHIP

- Erasmus Student Coordinator (ISEL) | 2018-2019
- Founder, EMBS Student Branch (ISEL) | 2018
- Chair, IEEE Student Branch (ISEL) | 2017-2018

## AWARDS & SCHOLARSHIPS

- FCT PhD Studentship | 2021-2025
- First in Class (PhD) | 2023
- Research Scholarship (ISEL) | 2021
- Merit Scholarship (Ministry of Education) | 2021
- First in Class (MSc Bio. Eng) | 2020

## ENGINEERING EXPERIENCE

- CEO – Memory Particle (Portugal) | 2017-2018
- Lead Mechanical – Brøvig (Norway) | 2016-2017
- Life Cycle Engineer – MH Wirth (Norway) | 2014-2015
- Project Engineer – Aker Solutions (Norway) | 2012-2014
- Mechanical Design – Siemens (Portugal) | 2011-2012

*Note: Senior engineering experience in large-scale, multidisciplinary and multinational industrial projects, involving coordination across diverse technical teams, communication with multiple stakeholder groups, and interface with executive-level decision makers.*